LABORATORY OBSERVATION

Course: Full Stack Web Development
Course Code: 23CM4121
Experiment No.: 3
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Date: 25-08-2026

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1. TITLE

Build an Interactive To-Do List Application with Dynamic DOM Addition and Deletion

2. AIM

To design and implement a responsive To-Do List web application using HTML, CSS, and vanilla JavaScript that allows users to add tasks, toggle completion states, delete tasks, and dynamically update a status message when the list is empty.

3. OBJECTIVE

- To configure real-time task creation and list insertion in JS.
- To attach separate event handlers to dynamically spawned buttons.
- To toggle styles dynamically (classList.toggle) to apply strike-through formatting.
- To remove DOM elements (taskItem.remove()) when deleting tasks.
- To dynamically hide/show empty placeholders depending on item counts.

4. THEORY

Vanilla JS event delegation involves defining callback routines that execute upon specific interactions with DOM nodes. By using document.createElement(), list elements <li> are constructed, populated with action buttons, and inserted under list blocks.

Removing elements is performed using child.remove() which immediately drops the targeted node from the active DOM tree. Class toggles (classList.toggle('completed')) apply or remove text-decoration: line-through styles, shifting element colors dynamically to reflect task states.

5. ALGORITHM / PROCEDURE

1. Create an HTML file named todo_list.html.
2. Write CSS styling for centered containers, margins, button colors, and strike-through text formatting.
3. Define input fields, add buttons, list items, and status placeholders in HTML.
4. In JS, select the input field, add button, list parent, and empty text block.

5. Write the updateEmptyMessage() helper function checking list children length.
6. Attach a click handler to the Add Task button.
7. In the handler, check if the input is empty; if so, trigger an alert dialog.
8. Dynamically create <li>, text spans, Complete buttons, and Delete buttons.
9. Attach a click listener to the Complete button toggling the completed style class.
10. Attach a click listener to the Delete button removing the parent list item from the DOM.
11. Append child nodes to the list item, then append the list item to the main list.
12. Reset inputs, call updateEmptyMessage(), and test the flow.

6. PROGRAM

todo_list.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>My To-Do List</title>
    <style>
        body { font-family: Arial, sans-serif; background-color: lightblue; display: flex;
justify-content: center; align-items: center; min-height: 100vh; margin: 0; padding: 20px;
}
        .container { background-color: white; width: 400px; padding: 20px; border-radius:
12px; box-shadow: 0 5px 15px rgba(0, 0, 0, 0.2); }
        h1 { text-align: center; color: #333; margin-bottom: 20px; }
        .input-section { display: flex; gap: 10px; }
        #taskInput { flex: 1; padding: 10px; border: 1px solid #aaa; border-radius: 6px;
font-size: 14px; }
        #addTask { padding: 10px 15px; background-color: #2196f3; color: white; border:
none; border-radius: 6px; cursor: pointer; }
        #addTask:hover { background-color: #1976d2; }
        #taskList { list-style: none; padding: 0; margin-top: 20px; }
        .task { display: flex; align-items: center; justify-content: space-between;
background-color: #f1f1f1; margin-bottom: 10px; padding: 8px 10px; border-radius: 6px; }
        .task-text { flex: 1; font-size: 14px; }
        .completed { text-decoration: line-through; color: gray; }
        .task button { margin-left: 5px; padding: 5px 8px; border: none; border-radius:
5px; cursor: pointer; font-size: 12px; }
        .complete-btn { background-color: #4caf50; color: white; }
        .delete-btn { background-color: #f44336; color: white; }
        #emptyMessage { text-align: center; color: #777; font-size: 14px; }
    </style>
</head>
<body>
    <div class="container">
        <h1>My To-Do List</h1>
        <div class="input-section">
            <input type="text" id="taskInput" placeholder="Enter a task.............">
            <button id="addTask">Add Task</button>
        </div>
```

```
        <p id="emptyMessage">No tasks available.</p>
        <ul id="taskList"></ul>
    </div>
    <script>
        const taskInput = document.getElementById("taskInput");
        const addTask = document.getElementById("addTask");
        const taskList = document.getElementById("taskList");
        const emptyMessage = document.getElementById("emptyMessage");

        function updateEmptyMessage() {
            if (taskList.children.length === 0) {
                emptyMessage.style.display = "block";
            } else {
                emptyMessage.style.display = "none";
            }
        }

        function createAndAddTask(taskText, autoComplete=false) {
            const taskItem = document.createElement("li");
            taskItem.className = "task";
            const text = document.createElement("span");
            text.className = "task-text";
            text.textContent = taskText;

            const completeButton = document.createElement("button");
            completeButton.textContent = "Complete";
            completeButton.className = "complete-btn";

            completeButton.addEventListener("click", function () {
                text.classList.toggle("completed");
                if (text.classList.contains("completed")) {
                    completeButton.textContent = "Completed";
                } else {
                    completeButton.textContent = "Complete";
                }
            });

            const deleteButton = document.createElement("button");
            deleteButton.textContent = "Delete";
            deleteButton.className = "delete-btn";
            deleteButton.addEventListener("click", function () {
                taskItem.remove();
                updateEmptyMessage();
            });

            taskItem.appendChild(text);
            taskItem.appendChild(completeButton);
            taskItem.appendChild(deleteButton);
            taskList.appendChild(taskItem);

            if (autoComplete) {
                text.classList.add("completed");
```

```
                completeButton.textContent = "Completed";
            }
            updateEmptyMessage();
        }

        addTask.addEventListener("click", function () {
            const taskText = taskInput.value.trim();
            if (taskText === "") return;
            createAndAddTask(taskText);
            taskInput.value = "";
        });

        // Auto trigger tasks for screenshot
        if (window.location.search.includes("render=true")) {
            window.addEventListener('DOMContentLoaded', () => {
                createAndAddTask("Complete Full Stack Lab Experiment", true);
                createAndAddTask("Prepare for Viva Voce", false);
            });
        }
    </script>
</body>
</html>
```

7. CODE EXPLANATION

Code	Explanation
`taskInput.value.trim()`	Retrieves input value and deletes leading/trailing spaces.
`document.createElement('li')`	Generates a new list item node for a task.
`text.classList.toggle('completed')`	Toggles class completed to cross out text when complete is clicked.
`completeButton.textContent = ...`	Swaps the text label of the complete button between Complete and Completed.
`taskItem.remove()`	Removes the task item element from the document tree, deleting it.
`taskList.appendChild(taskItem)`	Inserts the list item containing text and action buttons into the list element.
`updateEmptyMessage()`	Shows or hides the empty task message based on the count of items in the list.
`addTask.addEventListener('click', ...)`	Attaches a callback listener to trigger task

	addition when clicking the button.

8. TEST CASES AND EXECUTION DATA

Test Case	Task Input	Action / Input	Expected Result	Actual Result	Status
TC-01	Learn Node.js	Click Add Task	Task added to list, empty status message hidden	Added successfully	**Pass**
TC-02	Learn Node.js	Click Complete	Task text crossed out, button changes to Completed	Struck-through correctly	**Pass**
TC-03	Learn Node.js	Re-click Completed	Strike-through cleared, button changes back to Complete	Cleared correctly	**Pass**
TC-04	Learn Node.js	Click Delete	Task deleted from list, empty status message displayed	Deleted successfully	**Pass**

Additional Validation Test

Test Case	Input	Expected Result	Status
TC-05	Click Add Task with empty input	Browser triggers alert dialog showing 'Please enter a task!'	**Pass**

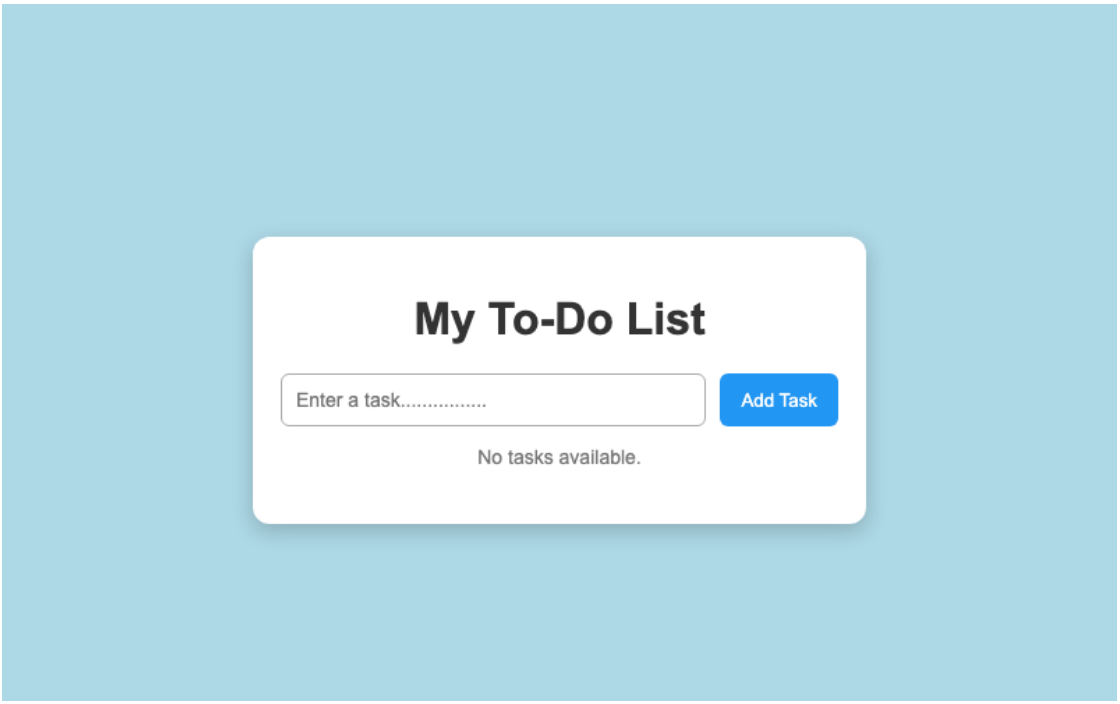
9. OUTPUT

Output 1: Execution Event Trace

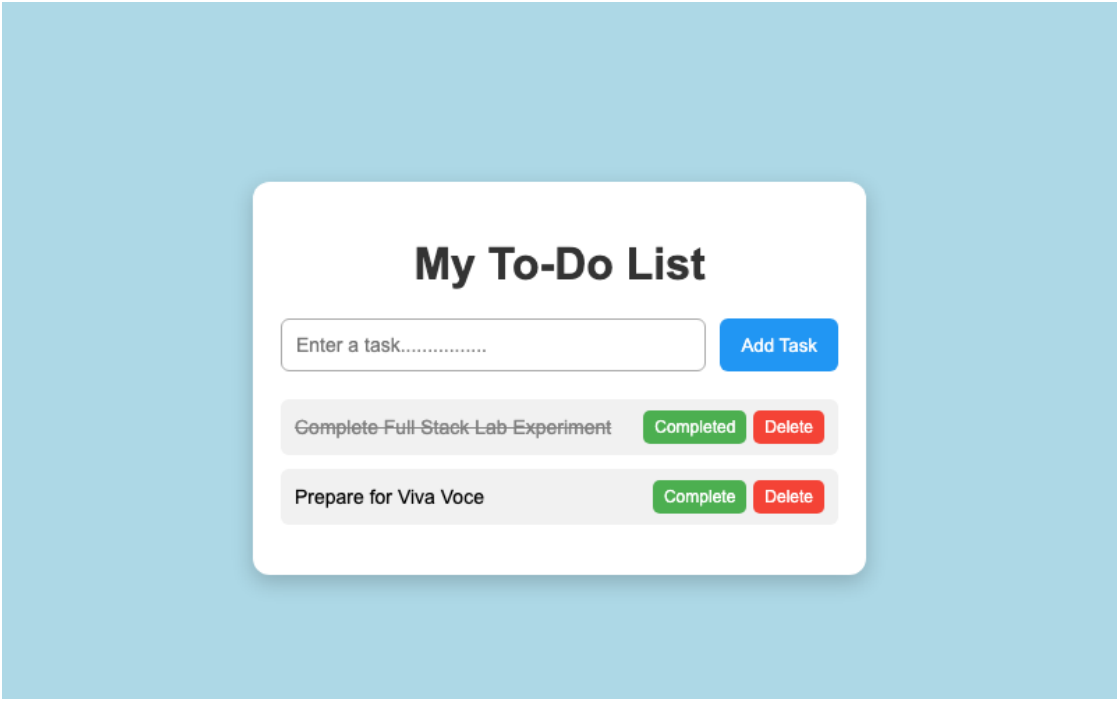
```
Loaded todo_list.html page
No tasks available message shown by default
Clicked Add Task with value 'Complete Full Stack Lab Experiment'
```

```
Clicked Complete -> task text crossed out
Clicked Delete -> task removed, placeholder shown
```

Output 2: Empty To-Do List Application



Output 3: Populated Tasks with Completion State



10. RESULT

Thus, an interactive To-Do List application with dynamic element insertion, toggleable strike-through completion status, dynamic task removals, and empty placeholder states was successfully built and tested.